

When Should an LLM Listen to Another LLM?

Predicting Beneficial Critique Acceptance in Multi-LLM Collaboration

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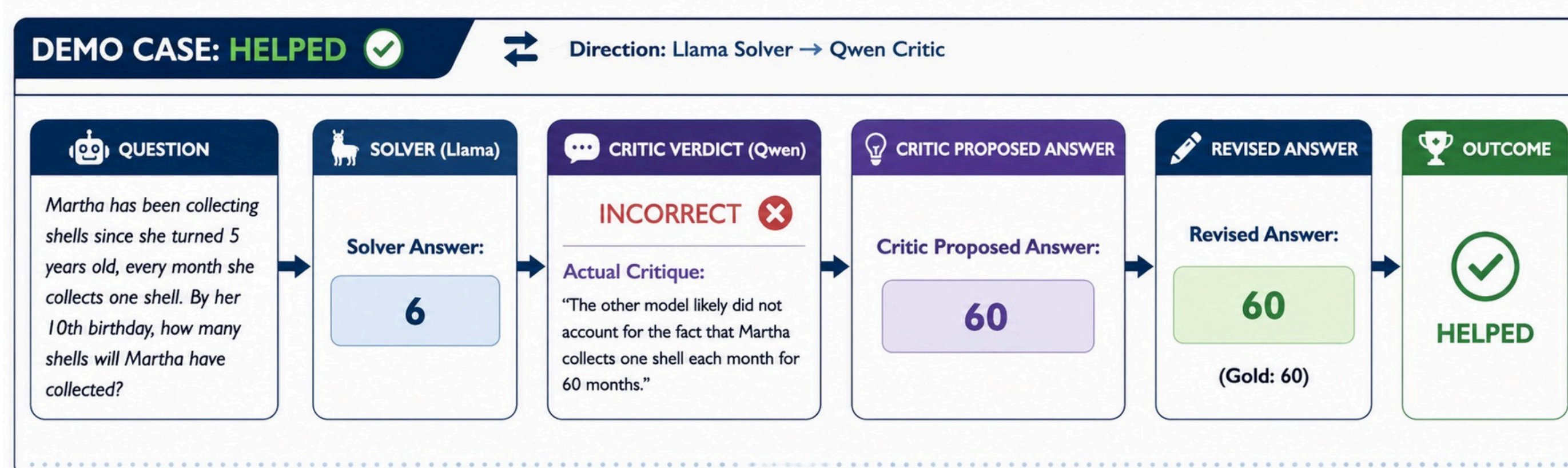
Background

- Multi-LLM critique is a double-edged tool:** external feedback can repair genuine errors, but accepting flawed critique causes models to abandon correct derivations.
- Self-refinement is unreliable** and assuming critique is uniformly beneficial degrades overall performance without selective decision gating.
- Core Question:** Can lightweight, pre-revision signals reliably predict when a solver LLM should accept external critique?

Objectives

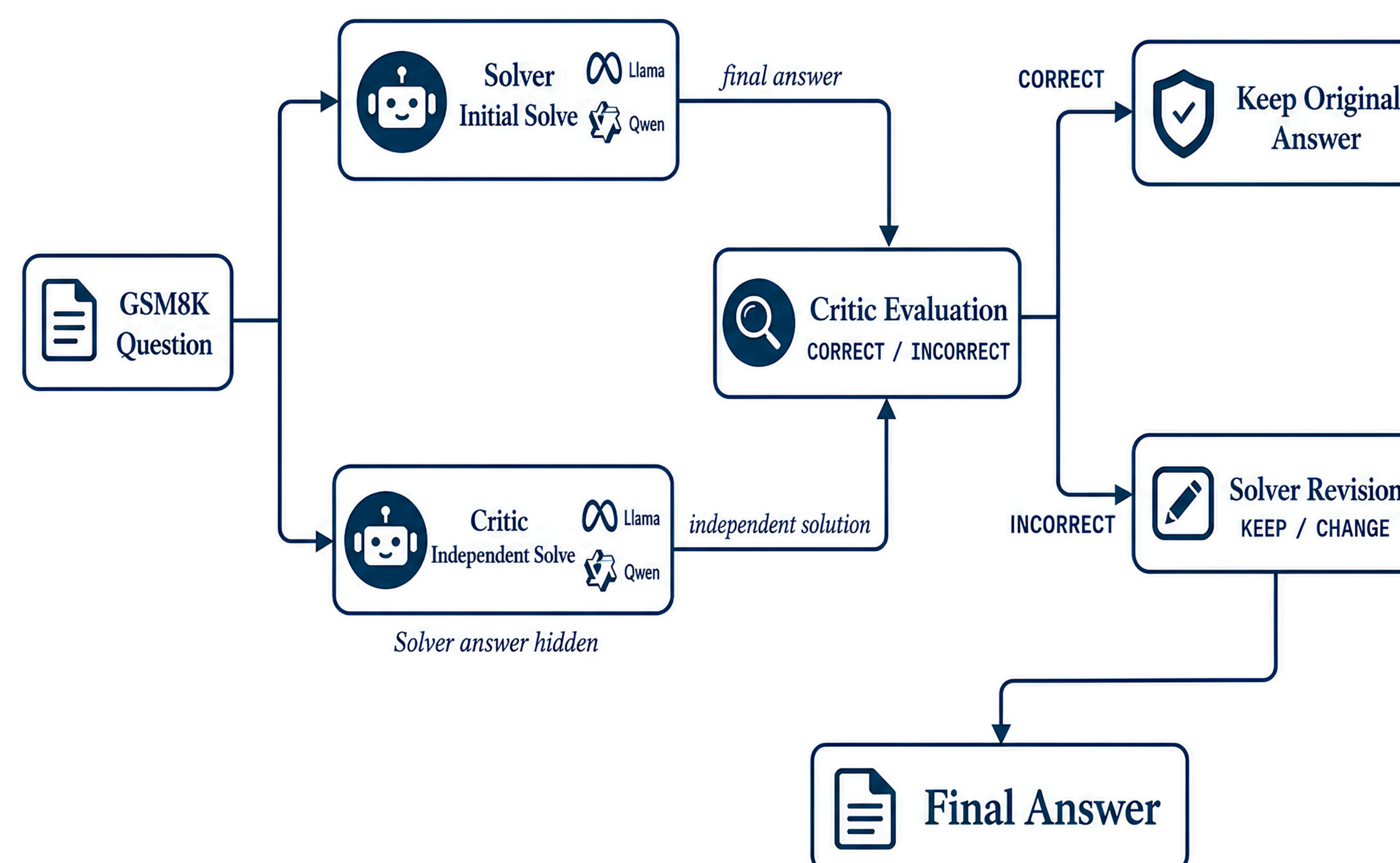
- Directional asymmetry:** Test whether critique benefit depends on solver–critic direction.
- Failure mechanism:** Investigate whether asymmetry is driven by differences in correction effectiveness or harmful critiques of correct answers.
- Predictive signals:** Compare directional-pairing features with critique-derived features for predicting beneficial revisions.

Demo Case

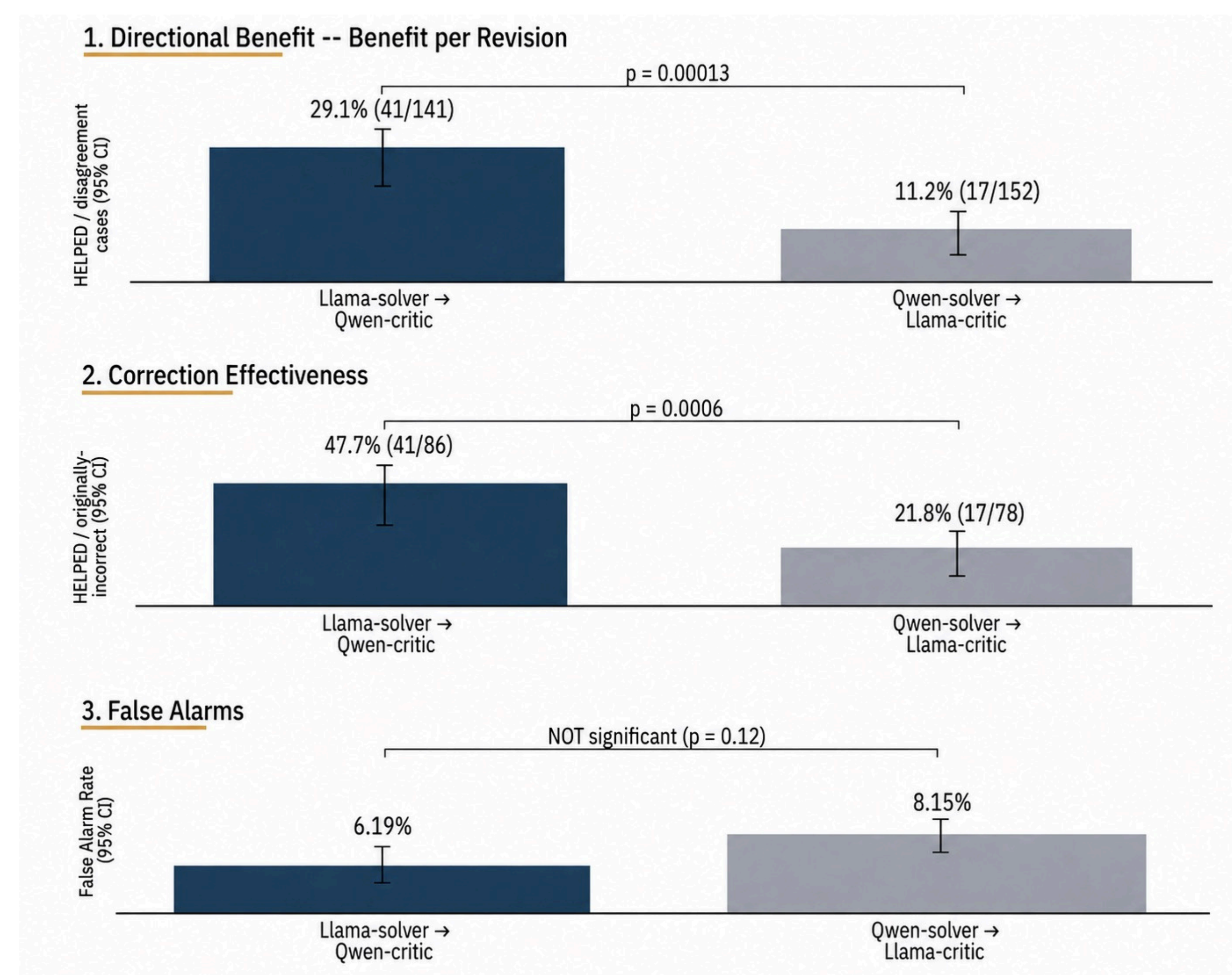


Methodology

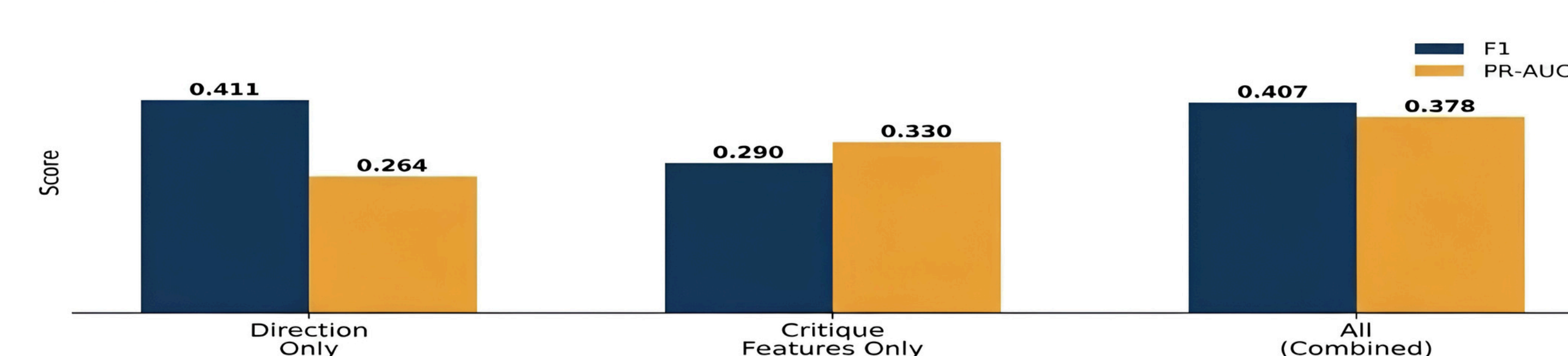
- Stage 1 | Solve:** Solver answers the problem; Critic solves it independently without seeing the Solver's output.
 - Stage 2 | Evaluate:** Critic compares the final answers and returns CORRECT / INCORRECT, an error description, and a proposed answer.
 - Stage 3 | Gate:** On disagreement, pre-revision features feed a balanced logistic regression model to predict whether revision will be beneficial.
 - Stage 4 | Validate:** 5-fold StratifiedGroupKFold grouped by question_id, followed by one-time evaluation on the frozen test set.
- Policies:** Never-accept · Adopt-critic · Always-accept · Learned policy · Oracle



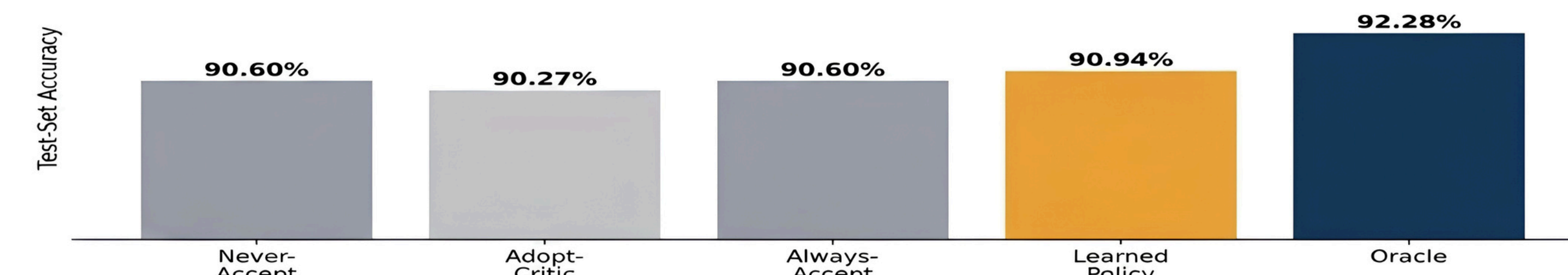
Results



4. 3-Way Feature Ablation (supporting analysis)



5. Decision Policy -- Frozen Test Set (n=298), 90.94% is preliminary (5 HELPED cases)



Future Directions

- Scale Evaluation Datasets:** Expand held out testing benchmarks to overcome statistical underpowering and validate directional reproducibility.
- Integrate LLM as a Judge:** Replace shallow text length proxies with automated critique quality scoring to capture nuanced reasoning errors.
- Expand Model Heterogeneity:** Benchmark diverse open-source and proprietary model families to test cross-architecture generalization.
- Broaden Beyond Disagreement:** Extend the selective pre-revision gating framework to evaluate model consensus and agreement scenarios.
- Inference Time Deployment:** Package the learned selective decision gate into a plug and play execution wrapper requiring zero model retraining.

Acknowledgements

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Project Webpage



GitHub

